



Write Order Module

AXI3 PROTOCOL

VLSI LAB

Student: Barak Ariely

Supervisor: Gil Stoler





Background & Architecture

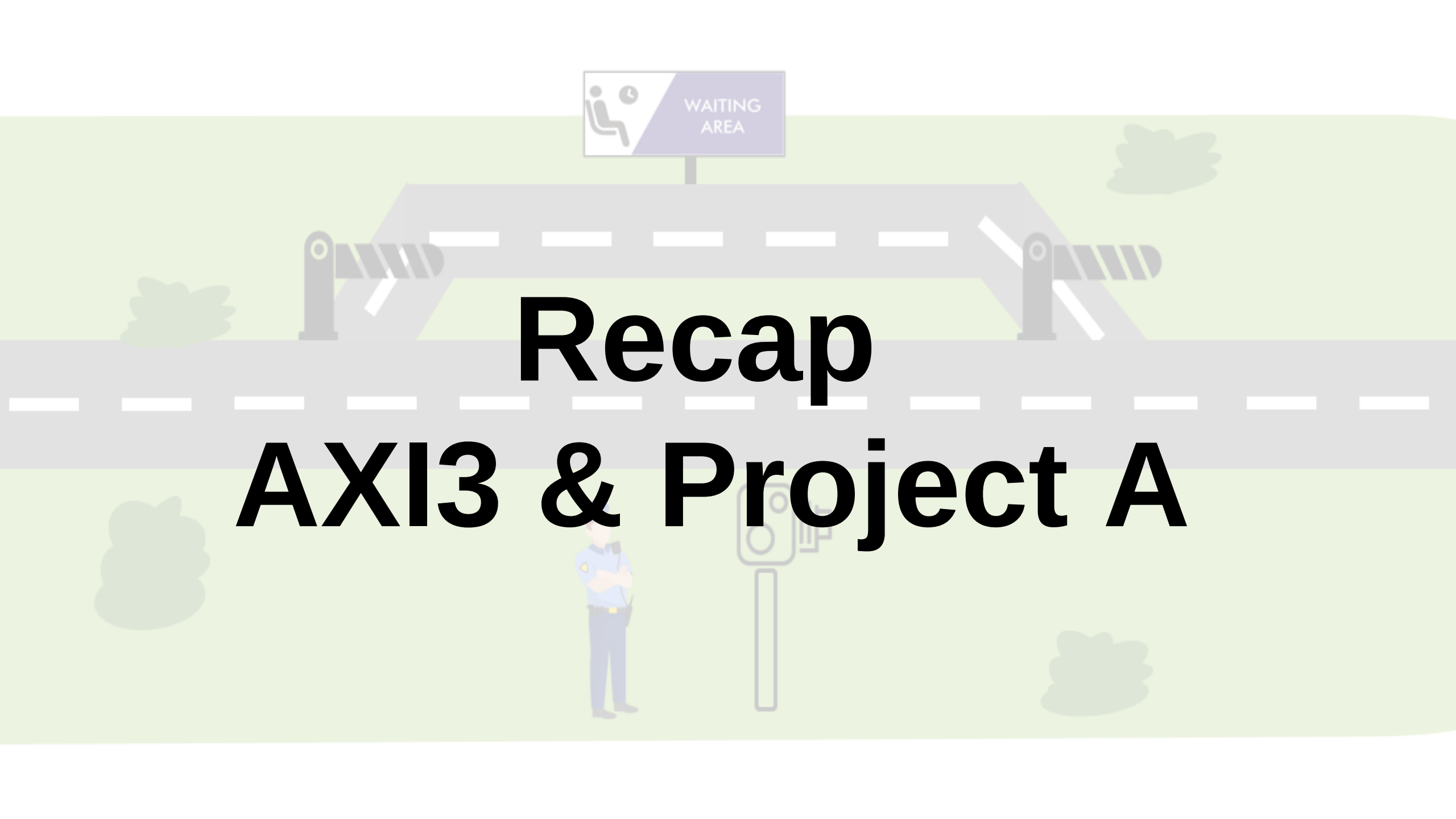
- Recap - AXI3 & Project A
- Goals
- Architecture Redesign
- Verification

Optimizations & Results

- Baseline & Frequency Ceiling
- Pipelining
- Architectural Rewrite
- Micro-Optimizations
- Results Summary

Overview



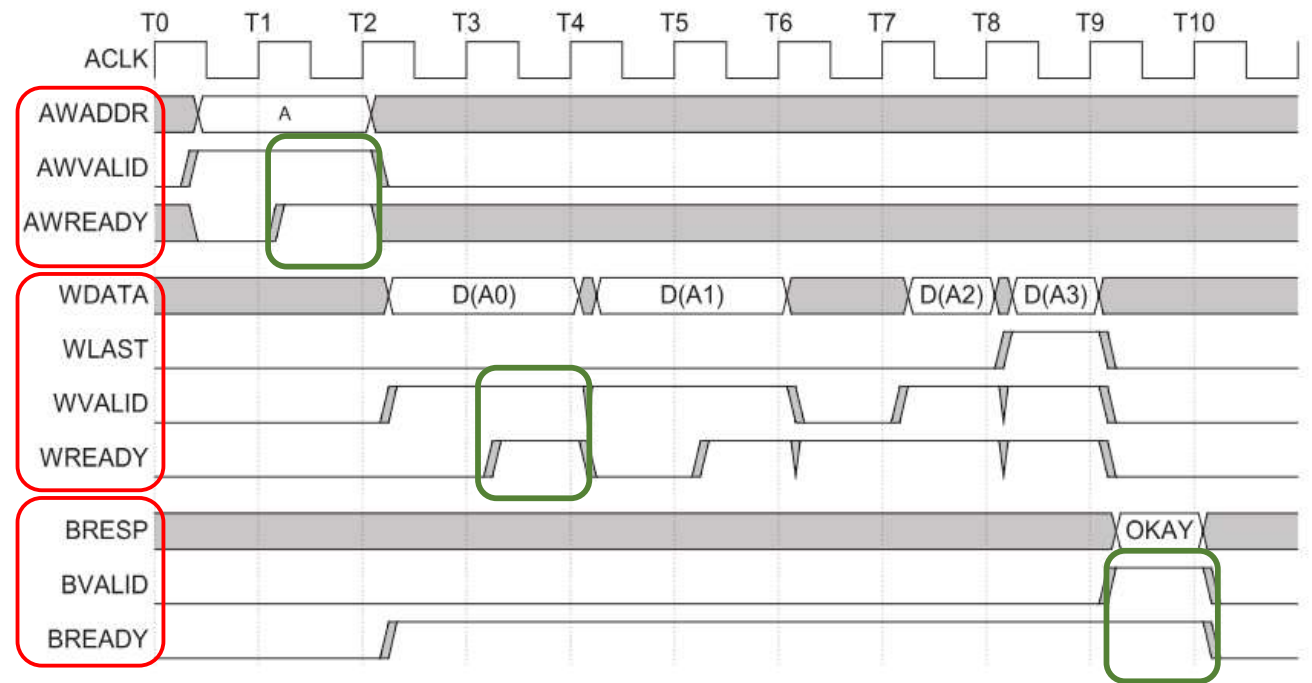


Recap AXI3 & Project A

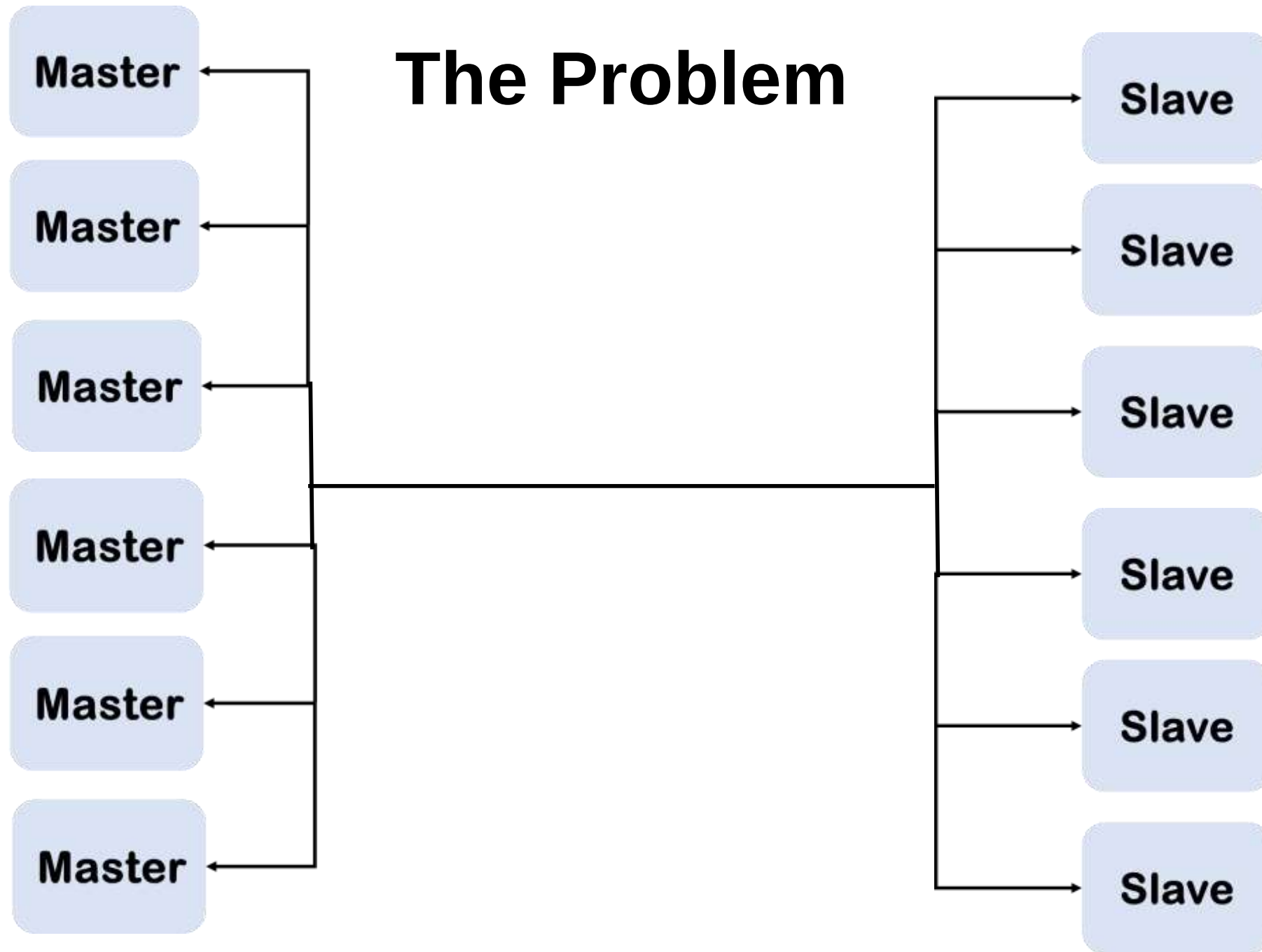
Recap – AXI3 Write Protocol

Serves most of ARM based fabrics/NICs.

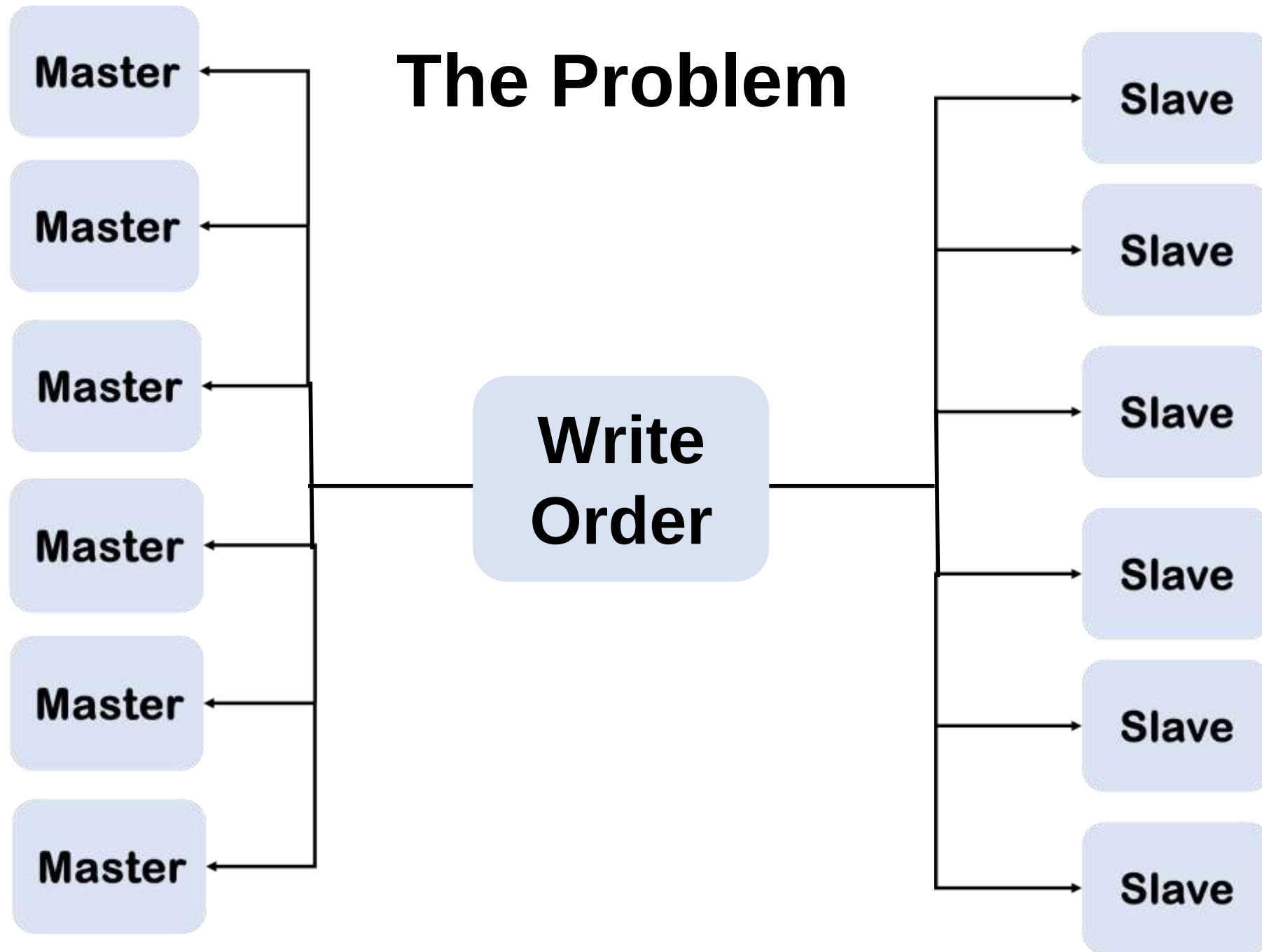
- The bus contains 3 Channels:
 - Address channel
 - Data channel
 - Response channel
- Transaction = Multiple transfers
- Handshake: Valid + ready bits at every channel



The Problem



The Problem

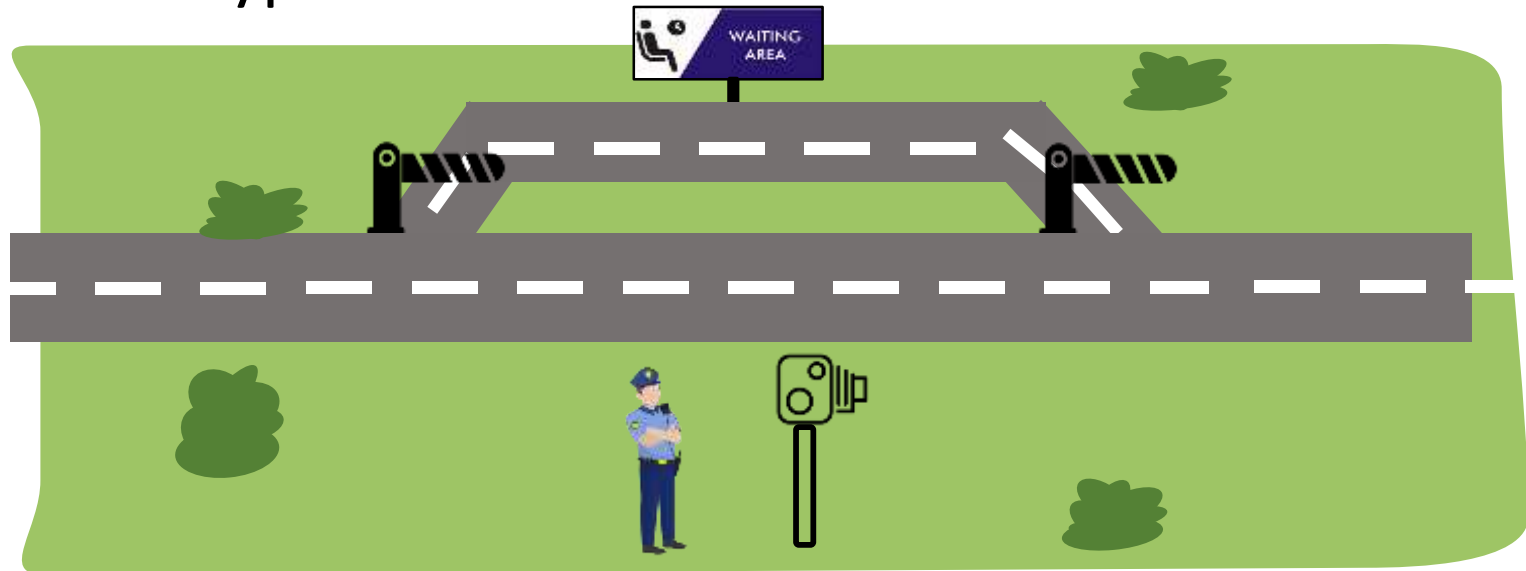


The Solution

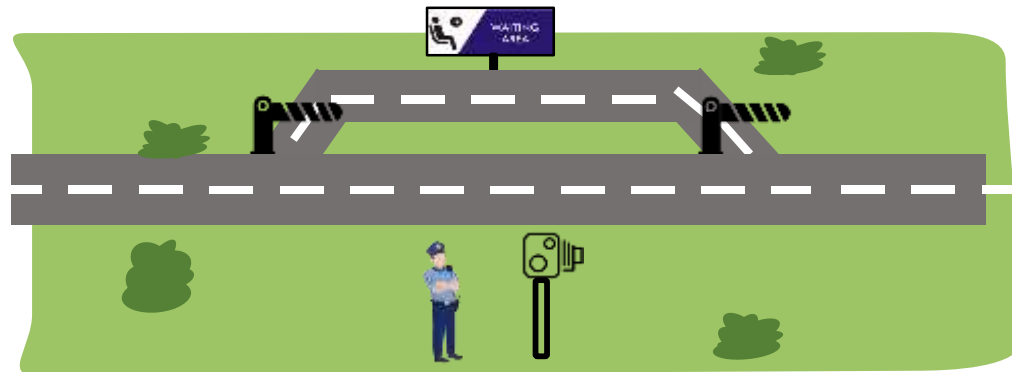
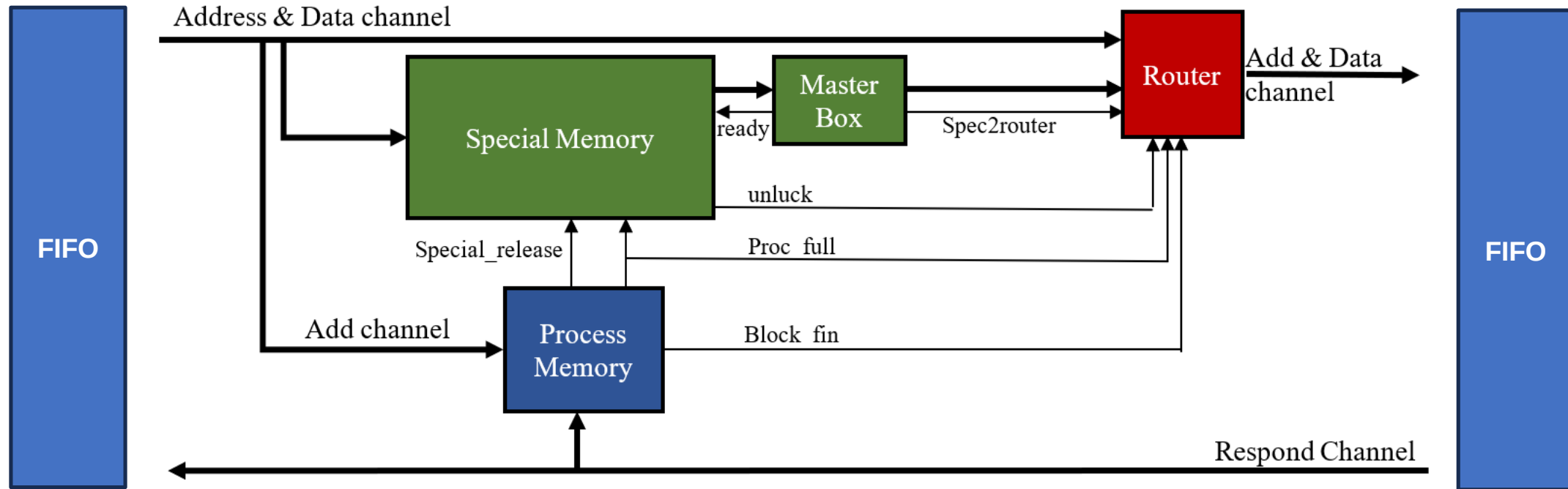
when optimizing the write path between a master and a slave/endpoint, one may find a reason for applying dependency on the passing transactions (data memory barriers), as a result of selected policies – special transactions.

What are the special transaction types?

1. Blocked transaction
2. Diverted transaction
3. Unlucky transaction



Project A Solution

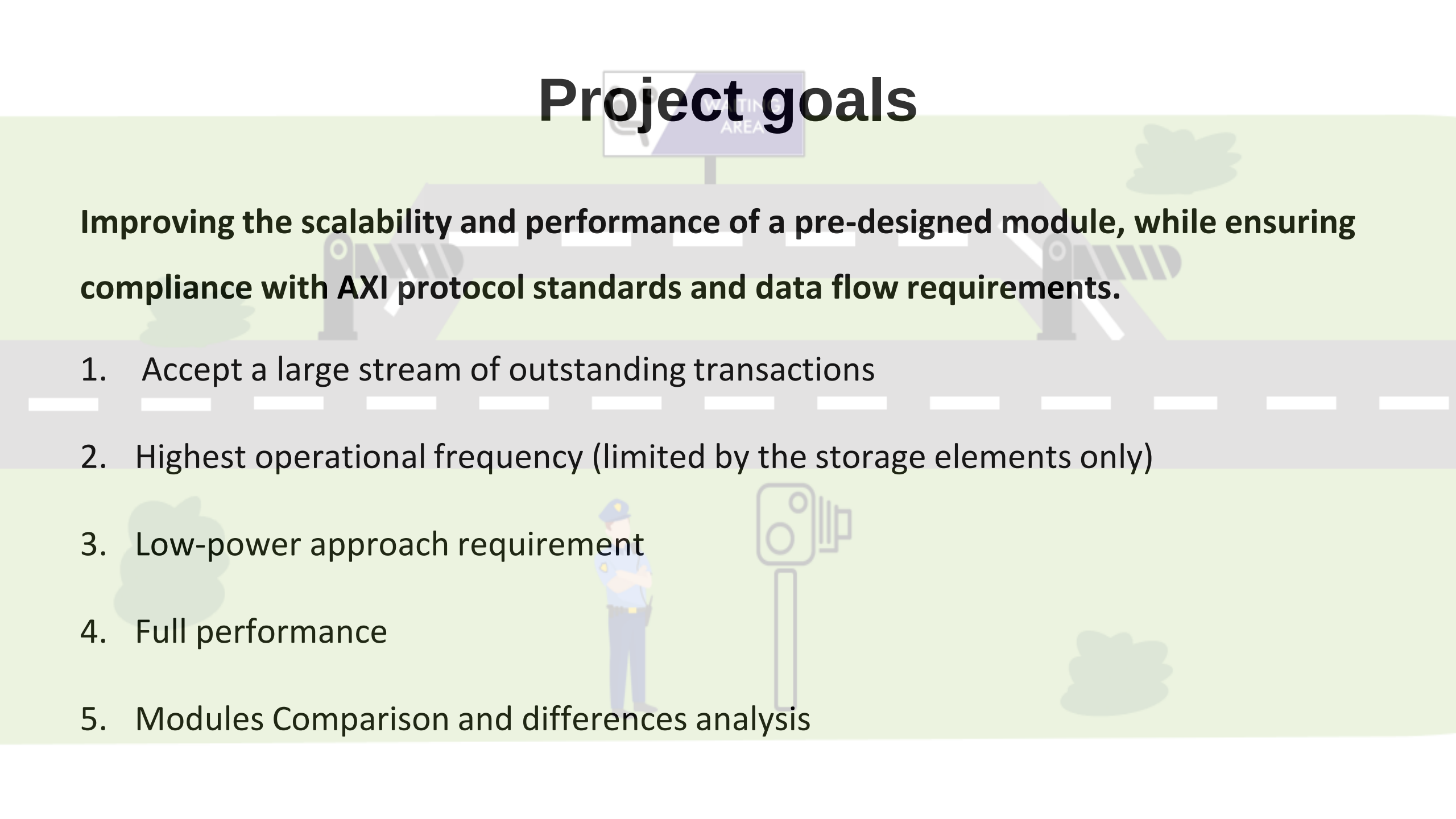




Goals

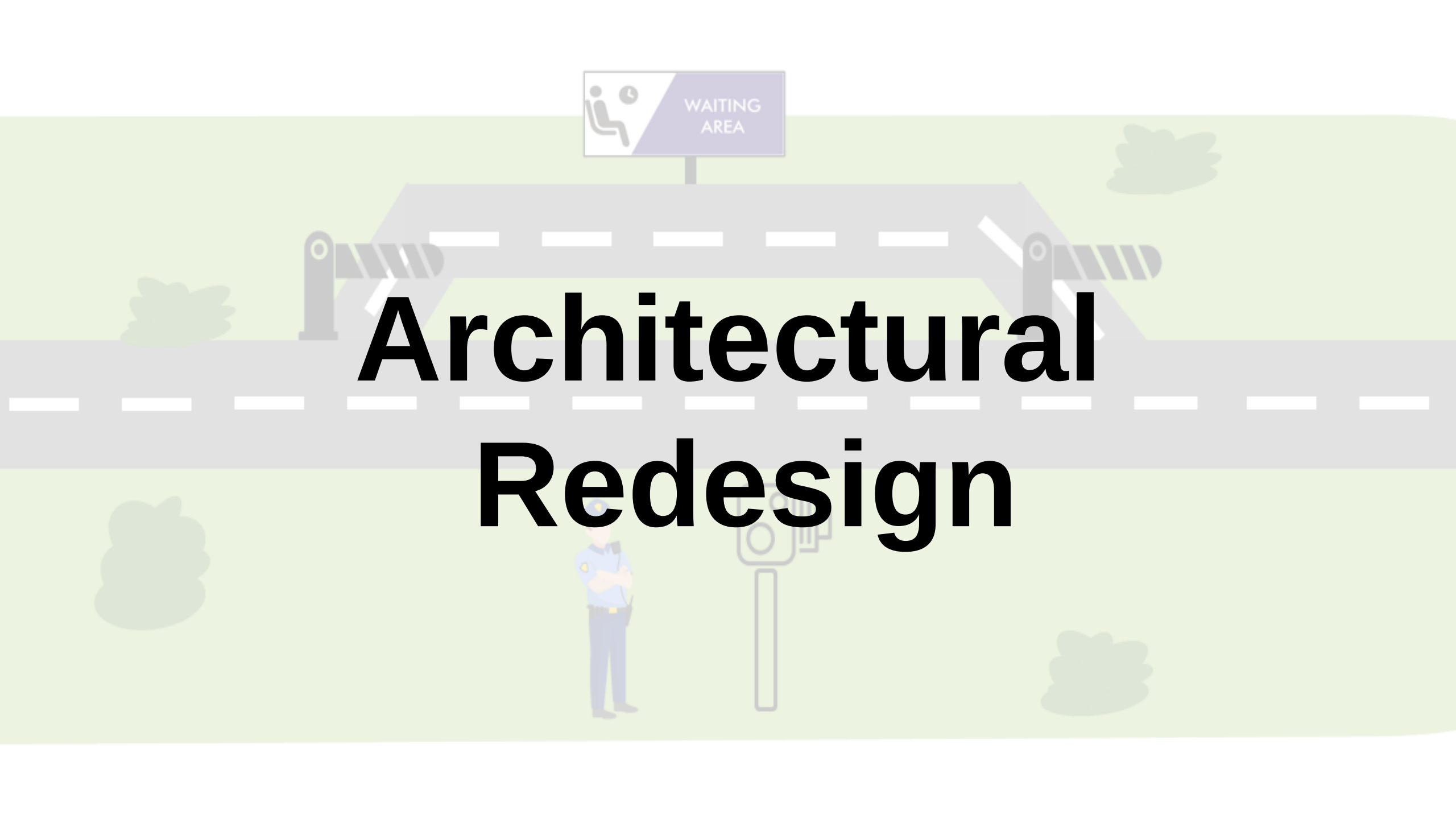


Project goals

The background of the slide features a stylized illustration of a road scene. At the top, a purple sign with a white arrow and the words 'WAITING AREA' is mounted on a pole. Below the sign, a road with white dashed lines leads towards a police officer standing on the grass. To the right of the officer is a speed camera on a tall pole. The scene is set against a light green background with some bushes.

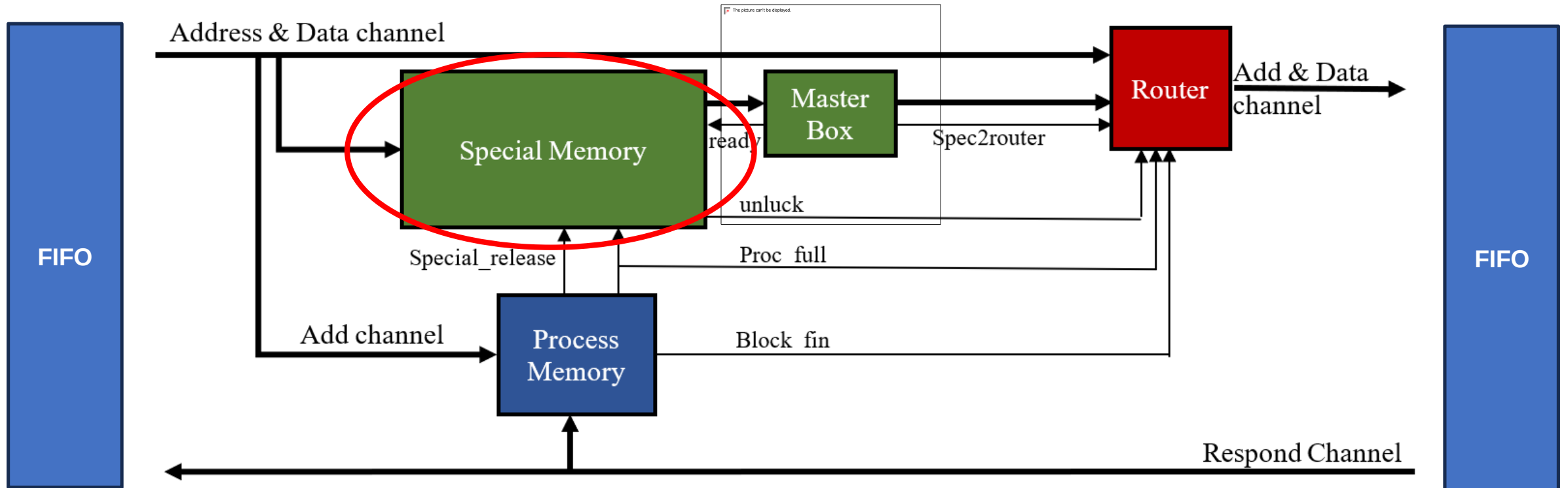
Improving the scalability and performance of a pre-designed module, while ensuring compliance with AXI protocol standards and data flow requirements.

1. Accept a large stream of outstanding transactions
2. Highest operational frequency (limited by the storage elements only)
3. Low-power approach requirement
4. Full performance
5. Modules Comparison and differences analysis

The background is a stylized illustration of a road scene. At the top, a purple sign with a white icon of a person sitting and a clock, and the text 'WAITING AREA', is mounted on a pole. Below the sign is a road with white dashed lines. A police officer in a blue uniform and cap stands on the grass to the left of the road. To the right of the officer is a white road sign on a pole. The scene is set against a light green background with some bushes.

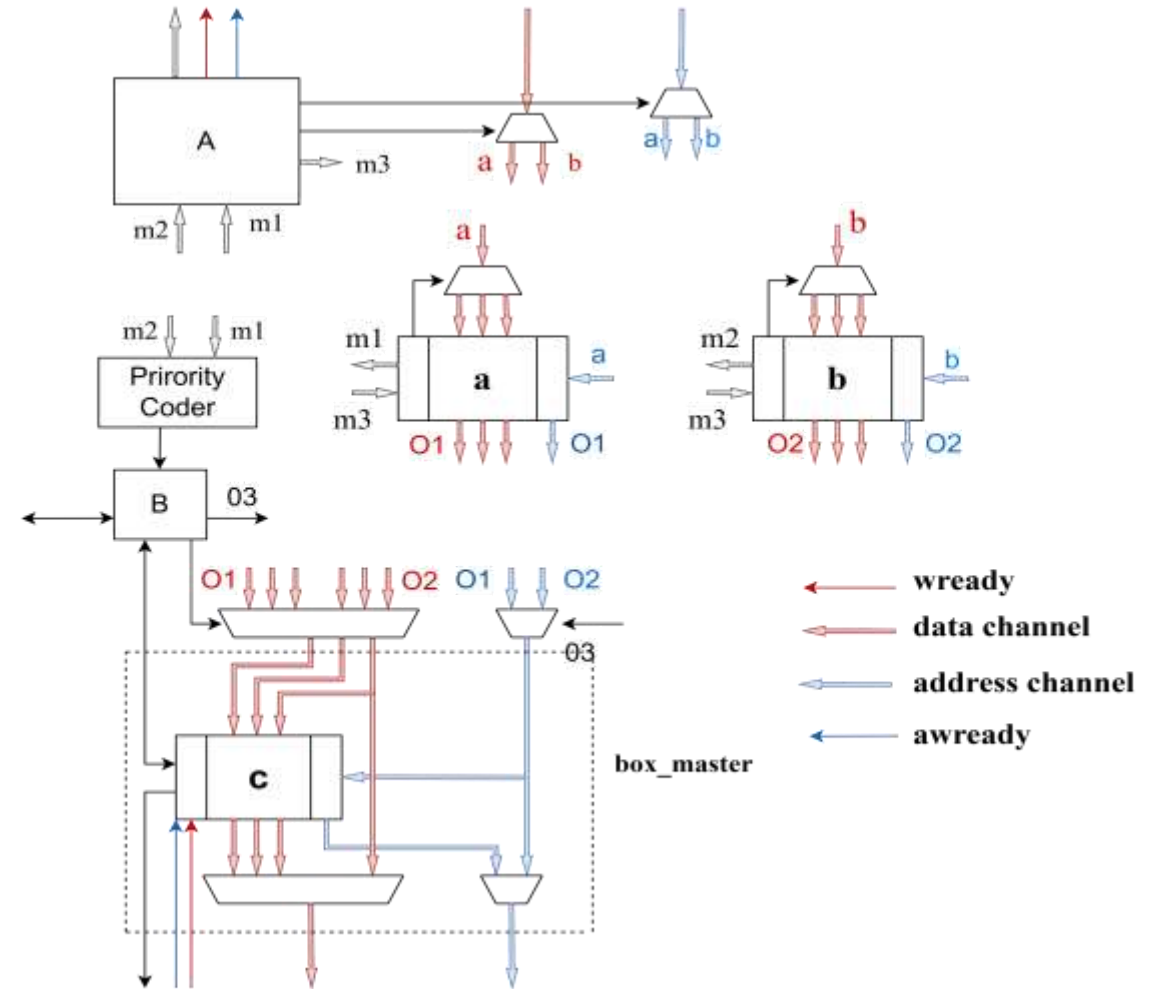
Architectural Redesign

Architectural Redesign



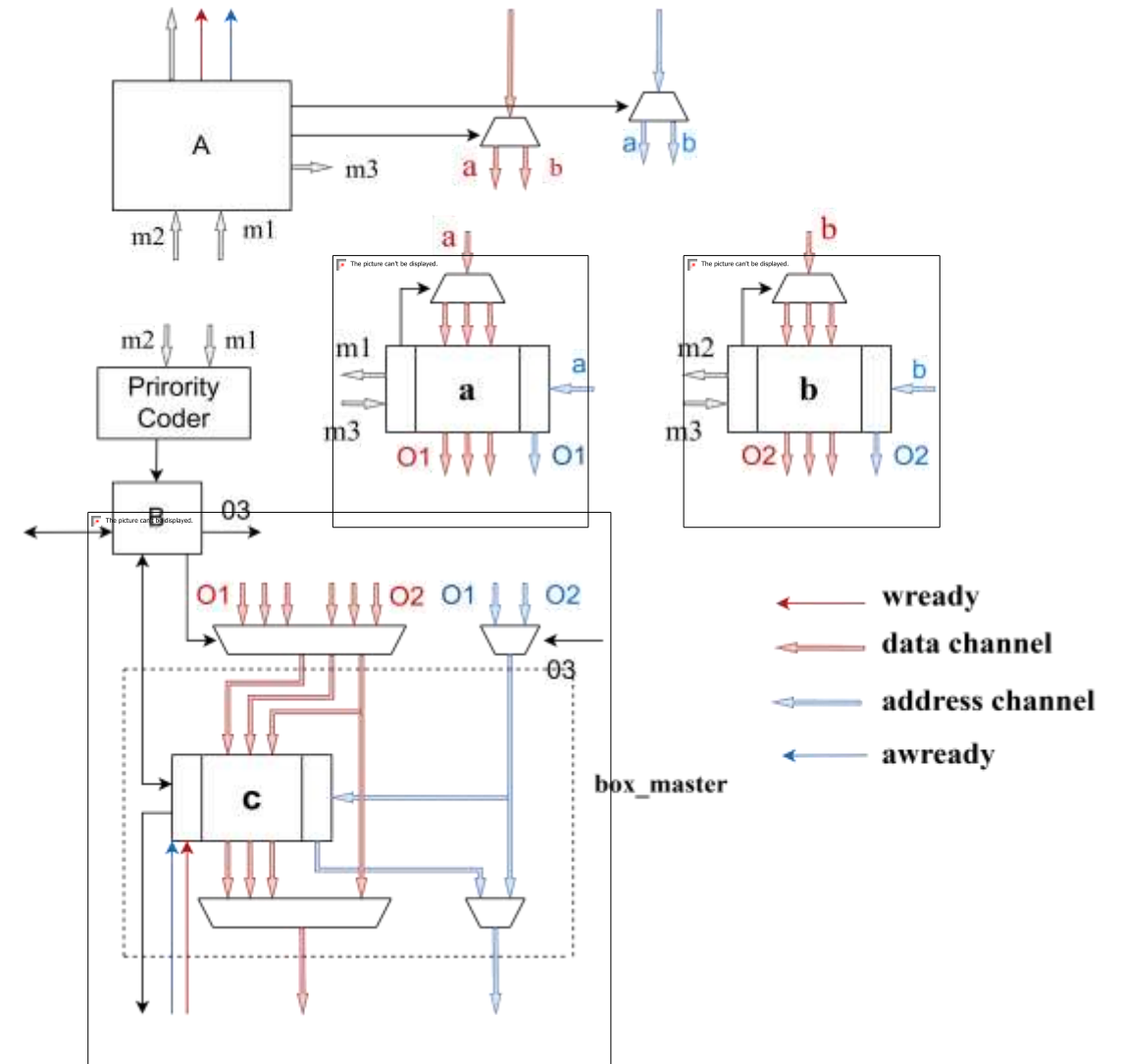
Special Memory

- **Shifted Array**
- **Static data → Moving Indexes**
Idea: Save dynamic power
- **Compare & Subtract deletion**
Transmit finished → Move all indexes

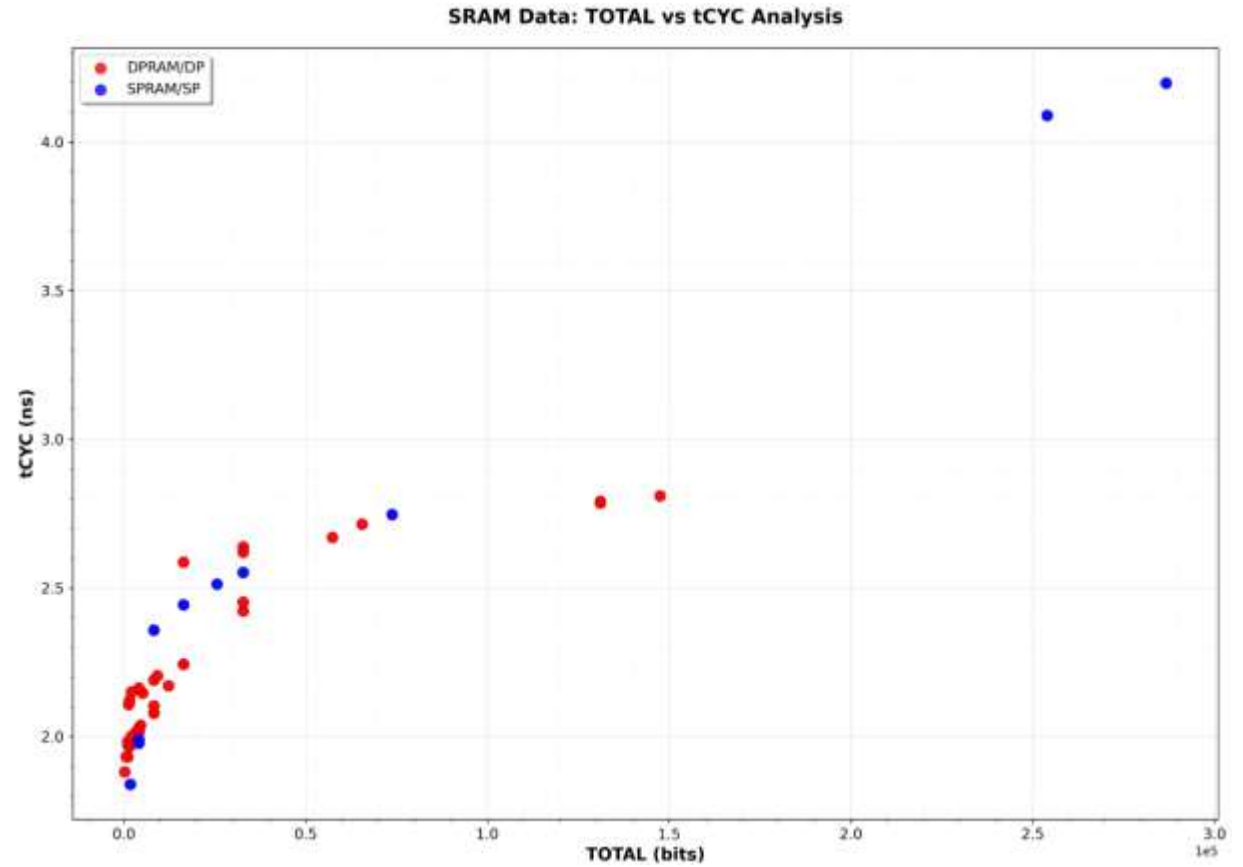
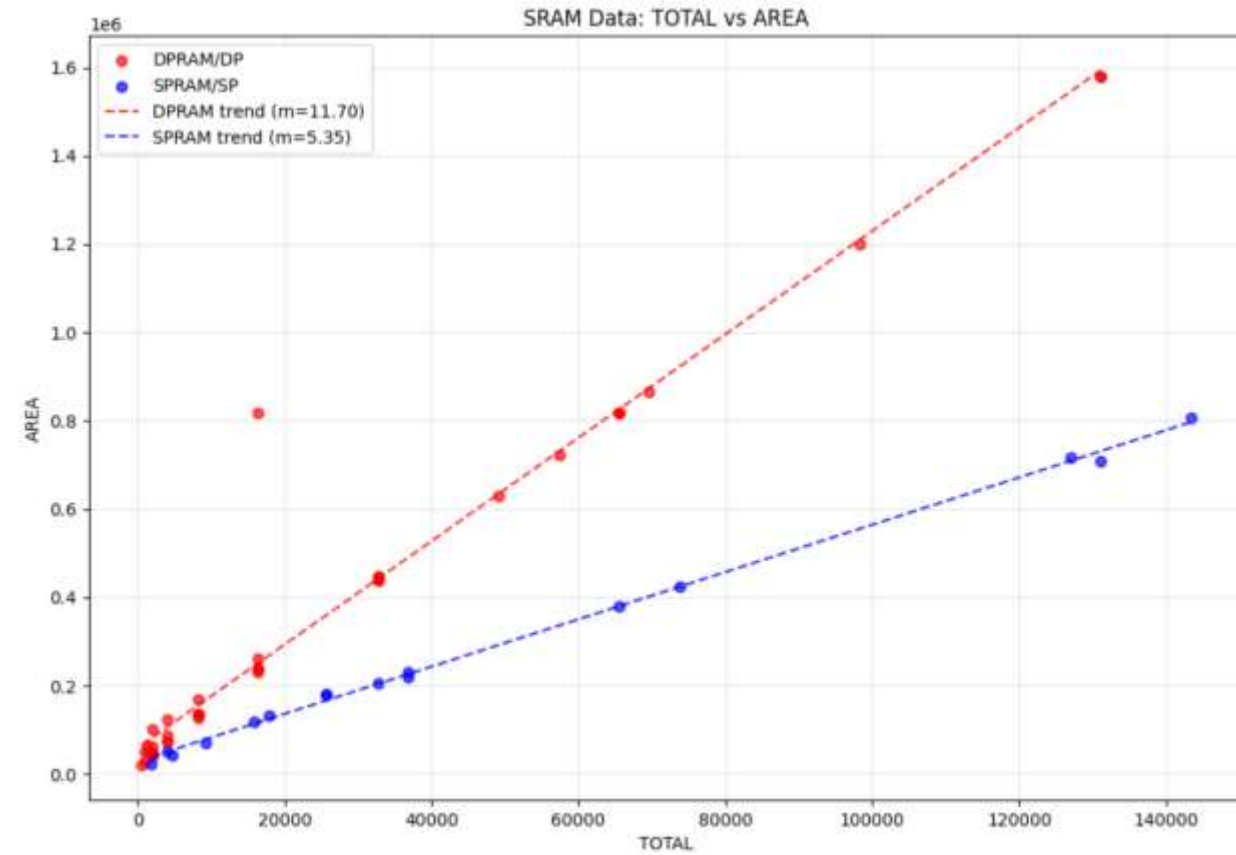


Special Memory

- Add **Parity Check**
- Replace **Registers** → **SRAM**
What size/type? What Architecture?
- Remove **box_master**

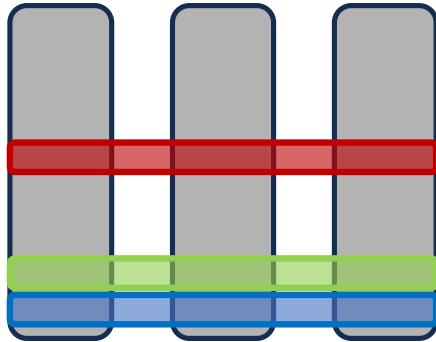


SRAM's Alternatives



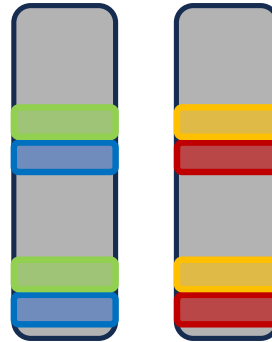
SRAM's Architecture Alternatives

(1)



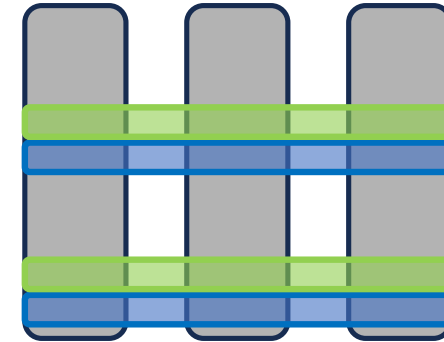
32x"SP16x72"

(2)



64x"SP8x72"

(3)

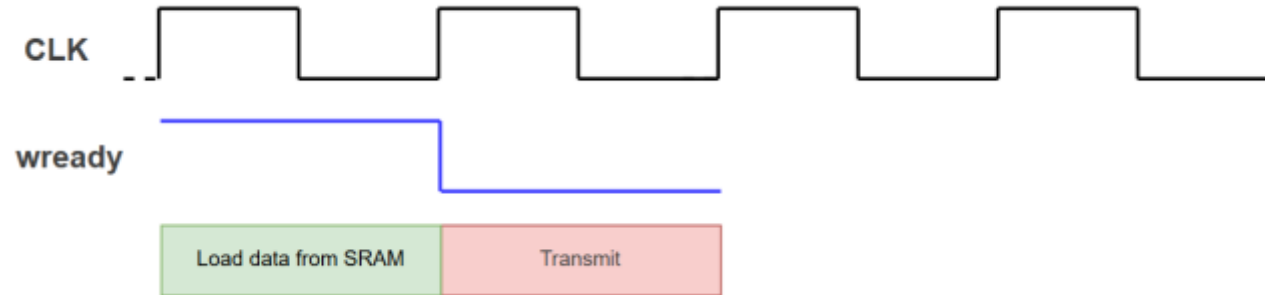


4x"DP128x72"

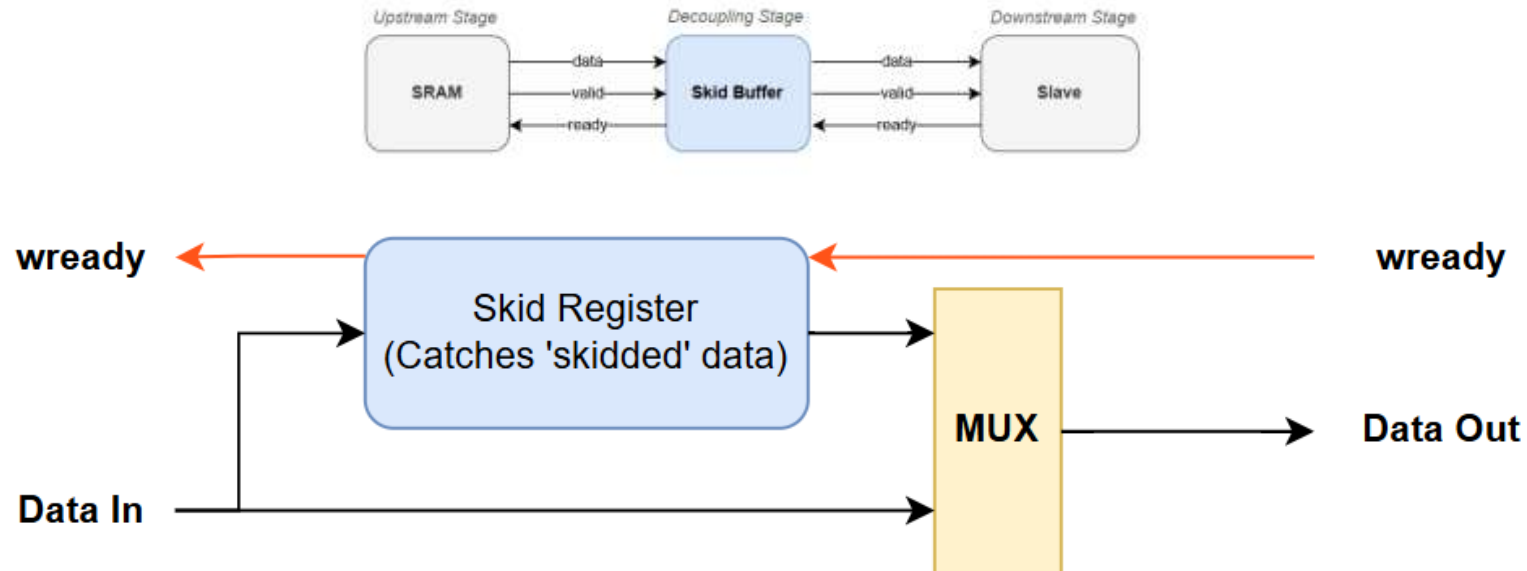
Power & Area **VS** "Scalable" Frequency **VS** Architectural Simplicity

Flow Control - Skid Buffers

Why



How



Verification

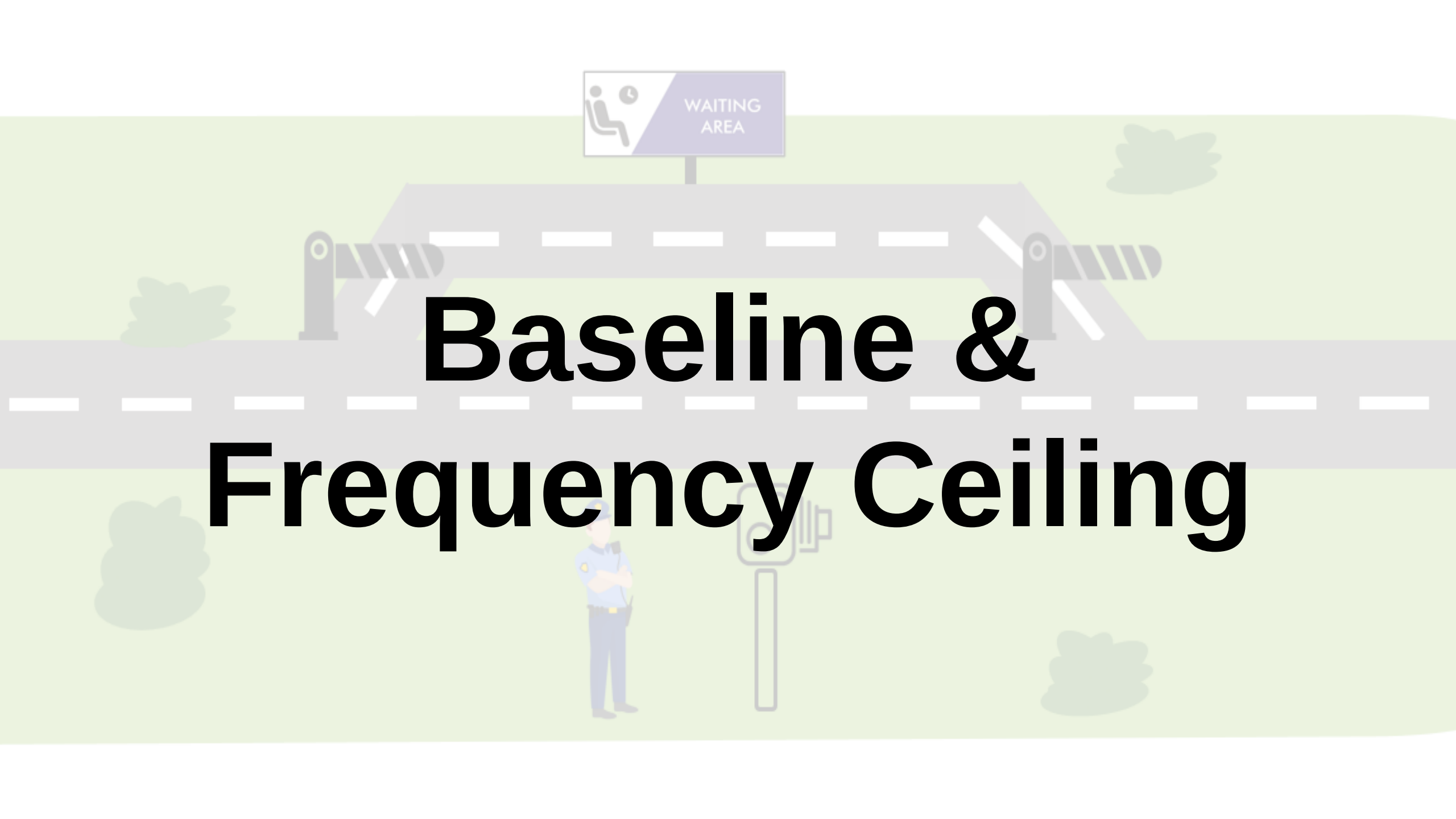
Challenge - Reach Testbench

SVA



Theory:

Minimal symptom coverage is often enough to uncover most bugs

The background is a stylized illustration. At the top, a purple sign with a white icon of a person sitting and a clock, and the text 'WAITING AREA', is mounted on a pole. Below the sign is a road with white dashed lines, flanked by two grey barriers with yellow and black diagonal stripes. In the foreground, a police officer in a blue uniform and cap stands with arms crossed. To the right of the officer is a white signpost with a circular sign. The ground is green with some grey bushes.

Baseline & Frequency Ceiling

Baseline

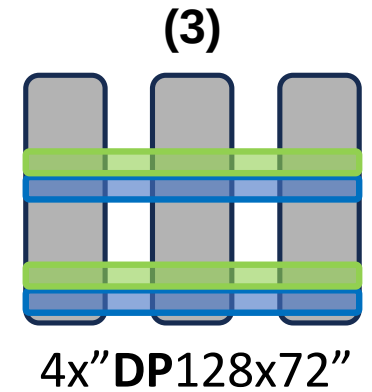
Synthesis results for cycle period **5.5 ns** (~181 MHz)

Metric	FF-based	SRAM-based	Improvement
Area	1,374,623 μm^2	967,222 μm^2	-30%
Dynamic Power	168.16 mW	108.89 mW	-35%
Timing (Slack)	0.000 ns	-0.329 ns	-

* Total Power $\approx$ Dynamic Power

SRAM-based baseline: Cycle period **5.9 ns** (~169 MHz)

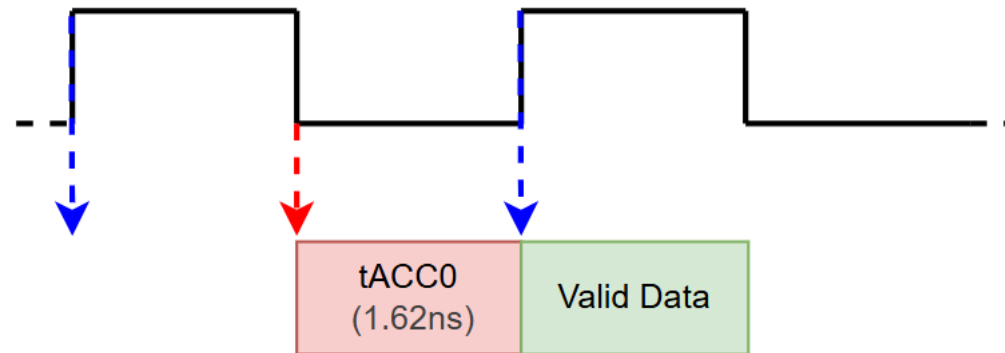
Frequency Ceiling



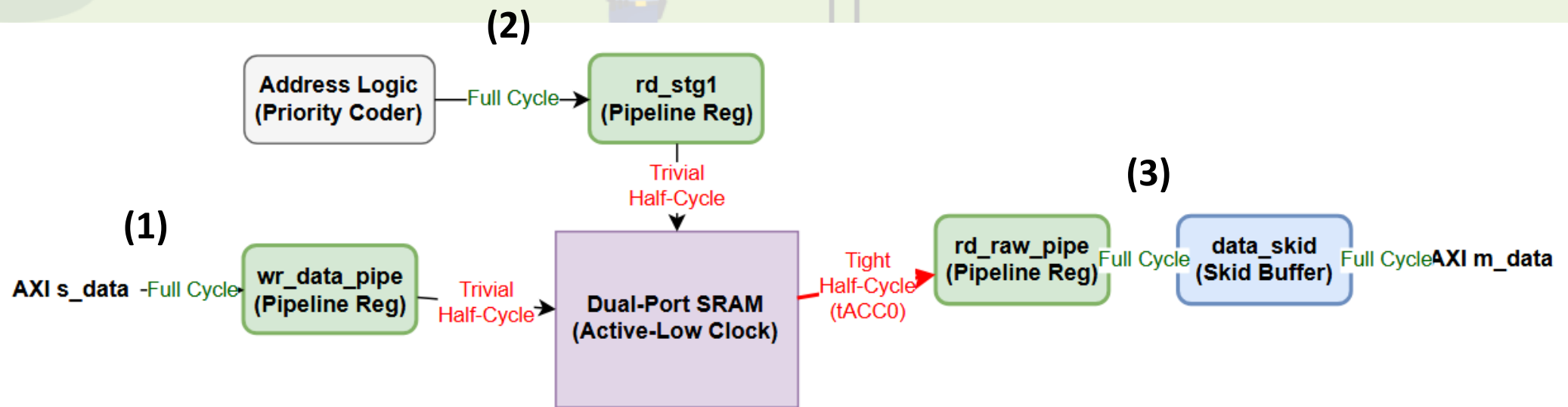
Architecture Choice: Strict single-cycle datapath

The Bottleneck (t_{ACC0}): SRAM internal read access == 1.62 ns

Absolute Limit: Period ≥ 3.24 ns (~ 310 MHz)

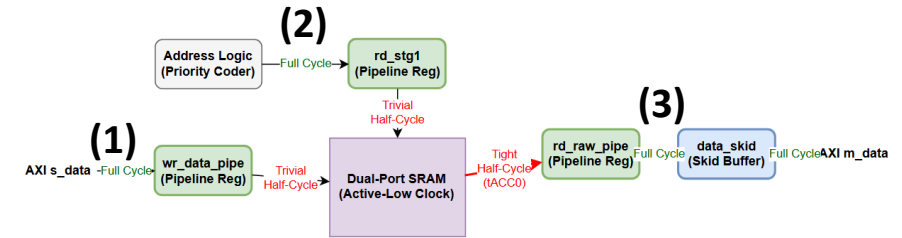


Pipelining

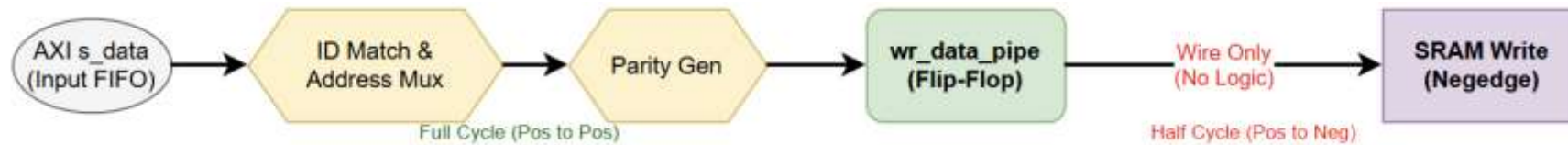


Pipelining

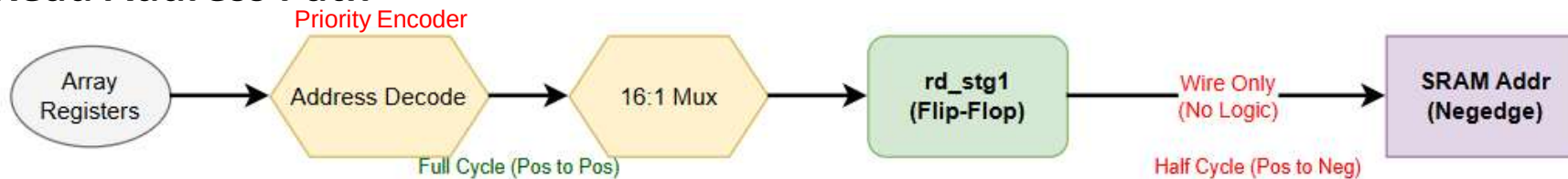
169 MHz (5.9 ns) → 222MHz (4.5 ns) +30%



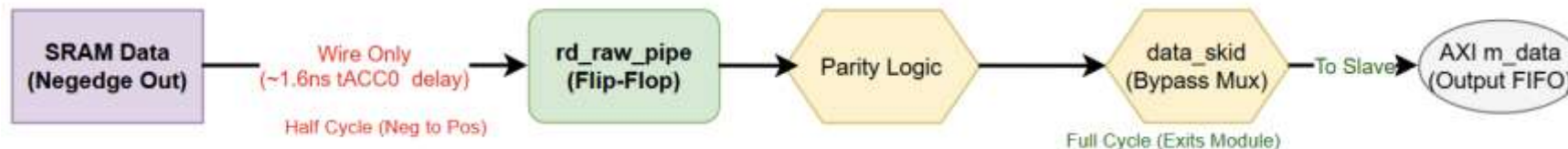
1. Write Data Path

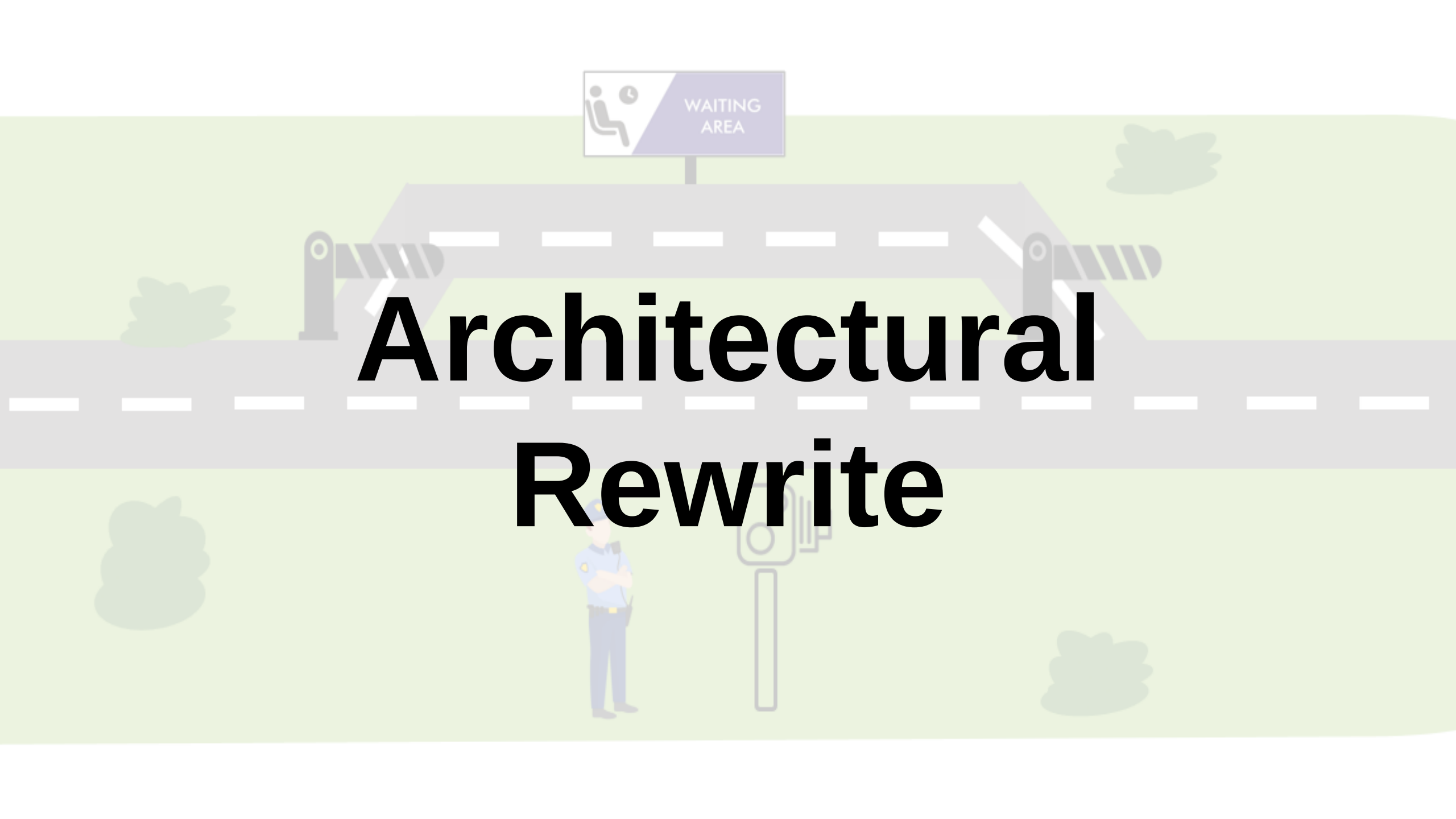


2. Read Address Path



3. Read Data Path



The background is a stylized illustration of a road scene. At the top, a purple sign with a white icon of a person sitting and a clock, and the text 'WAITING AREA', is mounted on a pole. Below the sign is a road with white dashed lines. A roadblock with two striped barriers is positioned across the road. In the foreground, a police officer in a blue uniform stands with arms crossed. To the right of the officer is a white signpost with a circular sign. The scene is set against a light green background with some bushes.

Architectural Rewrite

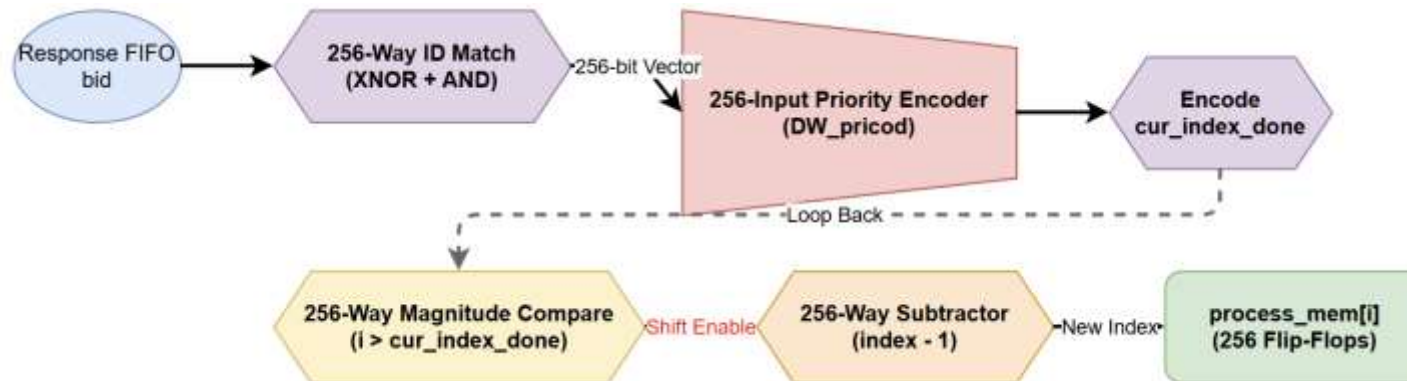
The Array Wall

Delete (Shift) Op

- Special Memory



- Process Memory



Let's find new **Solutions**

Or even better: Distill **Constraints**

Special Memory VS Process Memory





Entire Shifting + Priority Encoder



Bitmap

1
1
1

Age Matrix

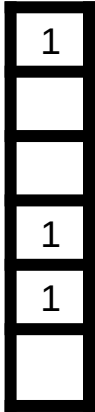
0						
1						
2						
3	1					
4	1			1		
5	1			1	1	

```
oldest[j] = mask[j] & ~|(age[j] & mask);
```

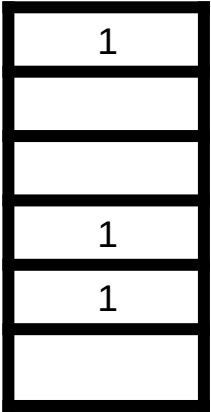
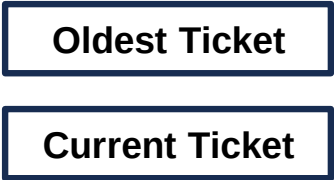
Metric	Before	After
Frequency	222 MHz (4.5 ns)	250 MHz (4 ns)
Logic Depth	$O(\log K)$ (Compare & Shift)	$O(1)$ Delete
Area	$O(K \log K)$	$O(K^2)$



Bitmap



Tickets



Entire Shifting + Priority Encoder



Metric	Before	After	
Area	1,105,839μm²	986,806μm²	-10.8%
FFs	15,053	13,134	-1,919

Metric	Before	After
Logic Depth	$O(\log N)$ (Compare & Shift)	$O(1)$ Delete
Area	$O(N \log N)$	$O(N \log K)$

The background is a stylized illustration of a road scene. At the top, a purple sign with a white icon of a person sitting and a clock, and the text 'WAITING AREA', is mounted on a pole. Below the sign is a road with white dashed lines. In the foreground, a police officer in a blue uniform stands with arms crossed. To the right of the officer is a speed camera on a tall pole. The scene is set against a light green background with some bushes.

Micro-Optimizations

- **Direct One-Hot Muxing**

Eliminated: Binary encoder + 16:1 mux to read metadata

if (one_hot[i]) idx |= 1
Value = vec[idx] $\longrightarrow$ value |= one_hot[i] & vec[i]

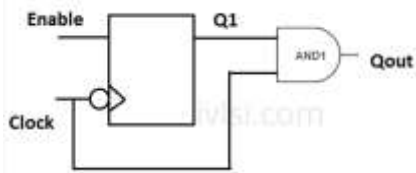
- **Parity Operand Isolation**

A 256-bit data bus requires a massive XOR gate tree
“Operand Isolation” with wstrb signal

- **Clock Gating**

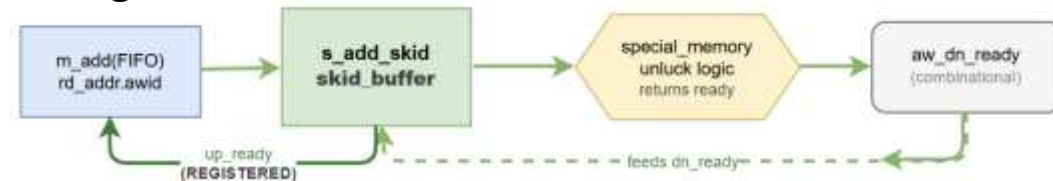
Using `compile_ultra -gate_clock` (ICG cells)

Idle slots are physically disconnected from the clock tree



- **FIFO to FIFO**

Breaking the Backward Path with a Skid Buffer



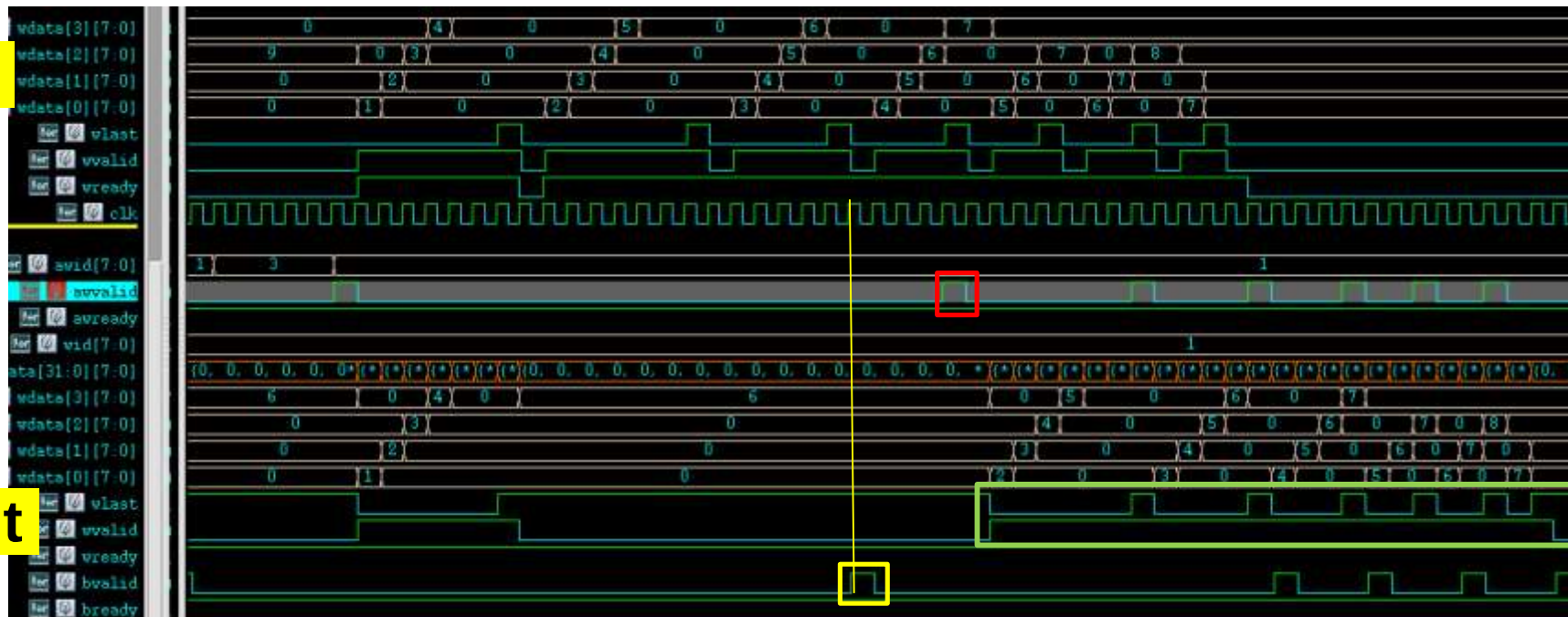


Summary & Conclusions



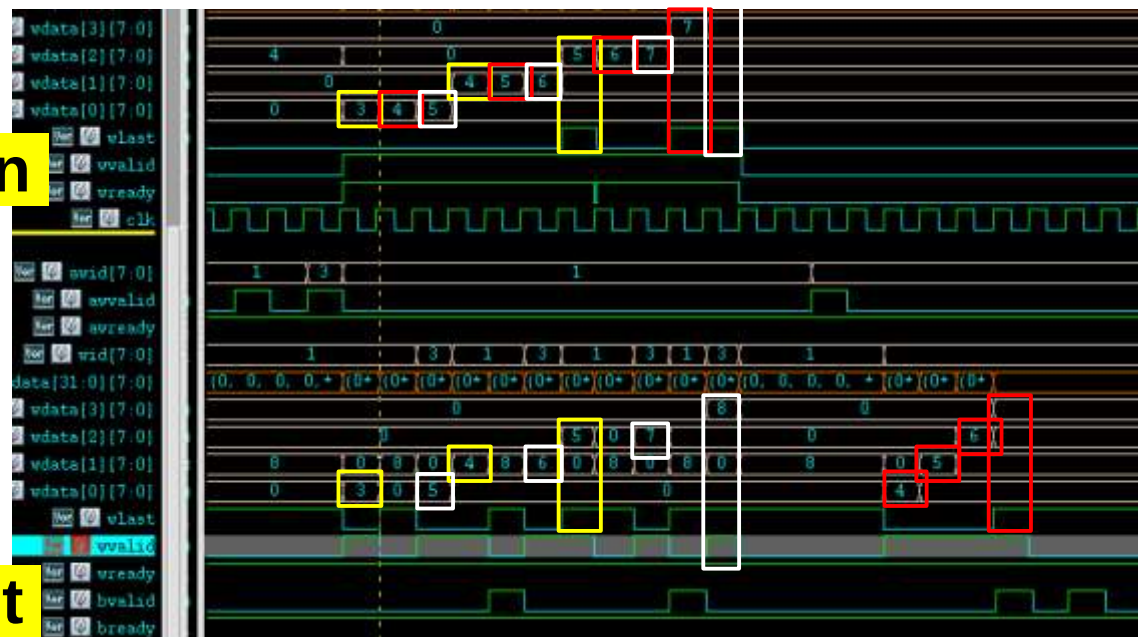
In

Out



In

Out



Startpoint: writetop/spec_mem/memory/a_ram_328x72_0
(falling edge-triggered flip-flop clocked by clk)
Endpoint: writetop/spec_mem/nd_raw_pipe/data_q_reg[17]
(rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max

Point	Incr	Path
clock clk (fall edge)	1.050	1.050
clock network delay (ideal)	0.000	1.050
writetop/spec_mem/memory/a_ram_328x72_0/c192 (dpram72x18.cb)	0.000	1.050
writetop/spec_mem/memory/a_ram_328x72_0/c2[64] (dpram72x18.cb)	1.431	1.281
writetop/spec_mem/nd_raw_pipe/data_q_reg[17]/0 (dff8_x18_svt)	0.000	1.281
data arrival time		1.281
clock clk (rise edge)	1.200	5.700
clock network delay (ideal)	0.000	5.700
writetop/spec_mem/nd_raw_pipe/data_q_reg[17]/0x (dff8_x18_svt)	0.000	5.700
library setup time	-0.044	5.700
data required time		5.700
data arrival time		1.281
slack (VIOLATED)		-0.025

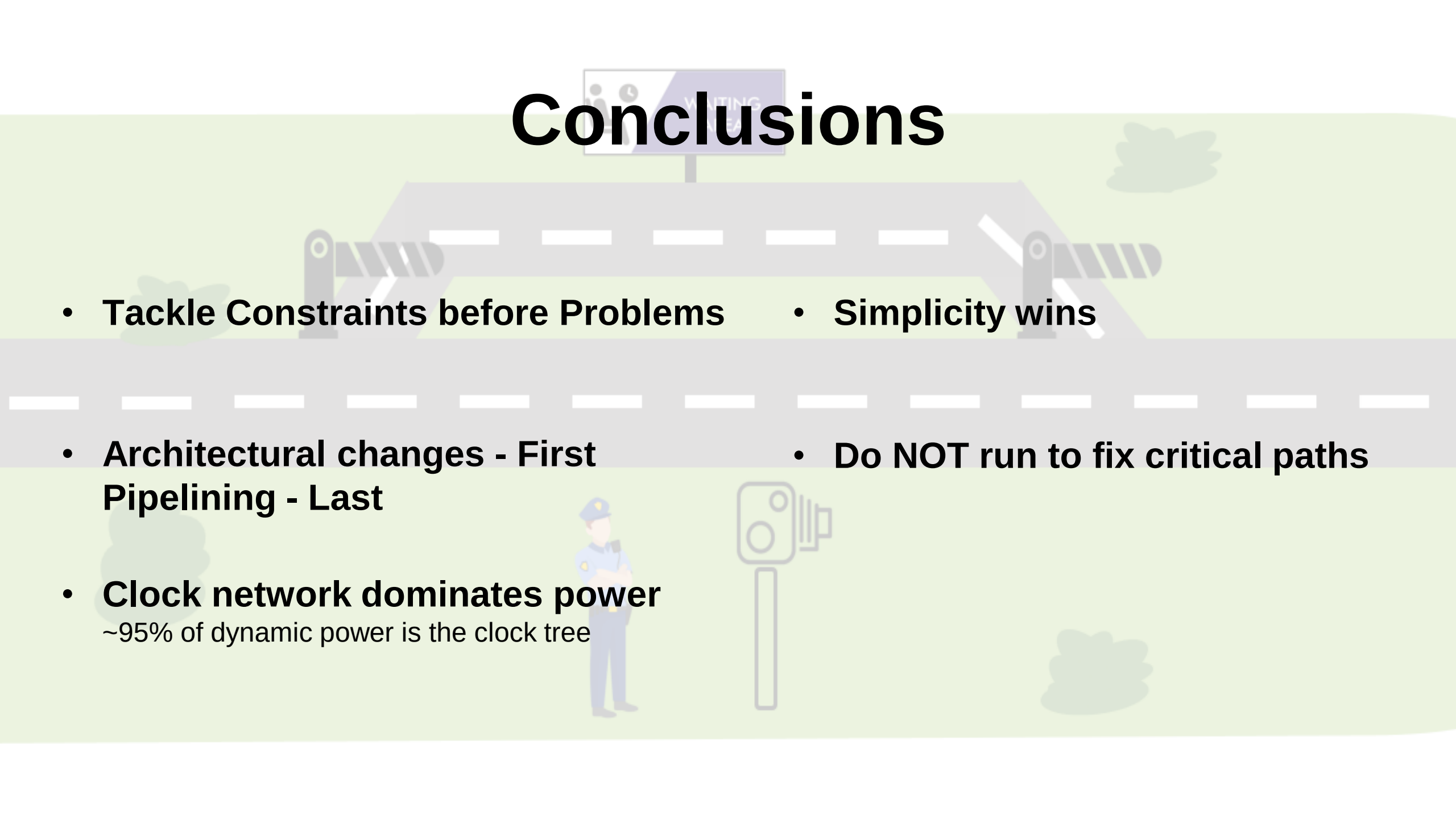
Summary

Metric	FF-based	SRAM-based	Final	vs FF	Vs SRAM
Area	1,374,623 μm^2	967,222 μm^2	986,806 μm^2	-30%	-28%
Dynamic Power	168.16 mW	108.89 mW	78.03 mW	-54%	-28%
Frequency	~180MHz (5.5ns)	~169 MHz (5.9ns)	~ 294MHz (3.4ns)	+64%	+74%

Scalability

Metric	Before		After (+SRAM)	
Memory Type	Process	special	Process	Special
Ops Depth	$O(\log N)$	$O(\log K)$	$O(1)$	$O(1)$
Area	$O(N \log N)$	$O(K \log K)$	$O(N \log K)$	$O(K^2)$

Conclusions

The background of the slide features a stylized illustration of a road scene. At the top, a purple sign with a clock icon and the word 'WAITING' is visible. Below it, a set of railroad tracks with crossing arms spans the width of the road. A police officer in a blue uniform stands on the grass to the left of the road. To the right of the officer, a speed camera on a pole is positioned. The road has white dashed lines, and there are green bushes on either side.

- **Tackle Constraints before Problems**
- **Simplicity wins**
- **Architectural changes - First**
Pipelining - Last
- **Do NOT run to fix critical paths**
- **Clock network dominates power**
~95% of dynamic power is the clock tree